Fundamental Physics II
Electricity – Chapter 2

For B1 Students
Academic year 2024-2025

Dr. Hoàng Thị Hồng Cẩm
Dept. of Advanced Materials Science and Nanotechnology
hoang-thi-hong.cam@usth.edu.vn

OVERVIEW

Chapter	Contents	Hours			Ref./Resources
		Lect.	Exr.	Prc.	
1	Coulomb's law*	2.0	1.0		
2	Electric field*	2.0	1.0		
3	Gauss' law	2.0	1.0		
4	Electric potential	2.0	1.0		
5	Magnetic force and magnetic field	2.0	1.0		
6	Biot-Savart's law and Ampere's law	2.0	1.0		
7	Induction	2.0	1.0		
8	Motion of charge in electric and magnetic fields	2.0	1.0		
		16	8		

Textbook:

[1] J. Parker, D. Halliday, R. Resnick, Fundamentals of physics 10th Edition, Wiley, IBSN 978-1-118-23072-5 (2007).

Assessment/ Evaluation		
	Attendance/Attitude	**10%**
	Exercise(s)	0%
	Practicals	0%
	Mid-term test	**30 %**
	Final exam	**60 %**

Chapter 2. Electric fields (2.0h)

2.1 The electric field

2.2 The electric field due to a charged particle

2.3 The electric field due to a dipole

2.4 The electric field due to a line of charge

2.5 The electric field due to a charged disk

2.6 A point charge in an electric field

2.7 A dipole in an electric field

List of Exercises:

In textbook "J. Parker, D. Halliday, R. Resnick, Fundamentals of physics 10th Edition, Wiley, IBSN 978-1-118-23072-5 (2007)."

22. 2, 15, 19, 28, 32, 35, 39, 50, 59 (Problems)

Chapter 2. Electric fields (2.0

2.1 The electric field

Two types of fields

The electric field

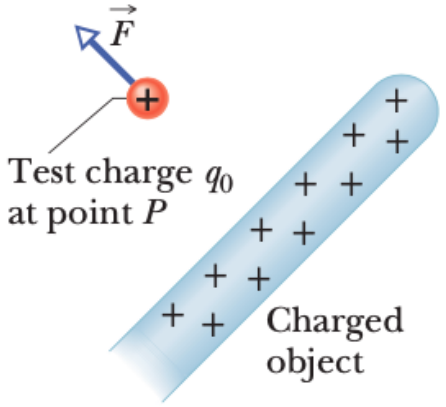
- ✓ Scalar fields (temperature field for an auditorium – the distribution of temperatures, pressure field in a swimming pool…): fields of scalar quantities, having only magnitudes and not directions.
- ✓ Vector fields : fields of vector quantities, having both magnitudes and directions.

Electric field at a point *P* (in SI unit: N/C*)*:

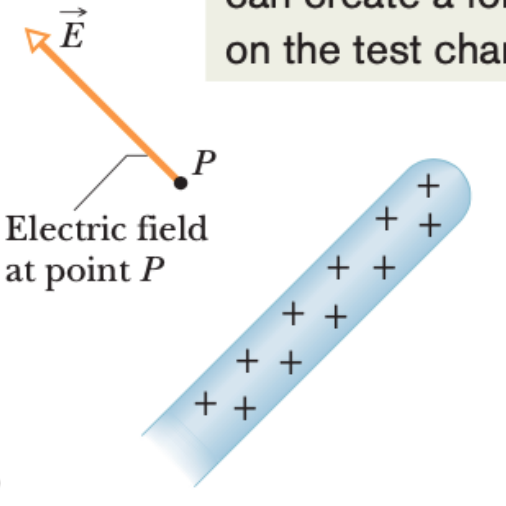
$$\vec{E} = \frac{\vec{F}}{q_0} \qquad (2.1)$$

q_0: the test charge (a small positive charge) at point *P*

$\vec{F}$: the electrostatic force that acts on the test charge



The rod sets up an electric field, which can create a force on the test charge.



(a) A positive test charge q_0 placed at point *P* near a charged object. An electrostatic force $\vec{F}$ acts on the test charge.

(b) The electric field $\vec{E}$ at point *P* produced by the charged object.

Chapter 2. Electric fields (2.0h)

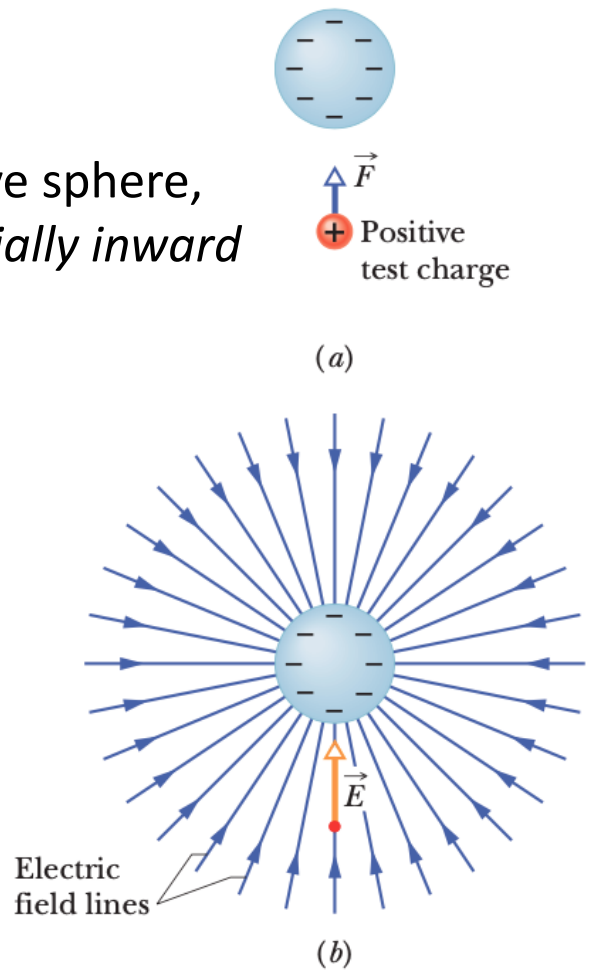
Electric field lines

✓ Introduced by Michael Faraday (19th century).

✓ The rules for drawing electric fields lines:

- At any point, the **electric field vector** must be **tangent to the electric field line** through that point and in the **same direction**.
- In a plane perpendicular to the field lines, the relative density of the lines represents the relative magnitude of the field there, with greater density for greater magnitude.

At every point around the negative sphere, an electric field vector points *radially inward toward* the sphere.



➢ Electric field lines extend *away from* positive charge (where they originate) and *toward* negative charge (where they terminate).

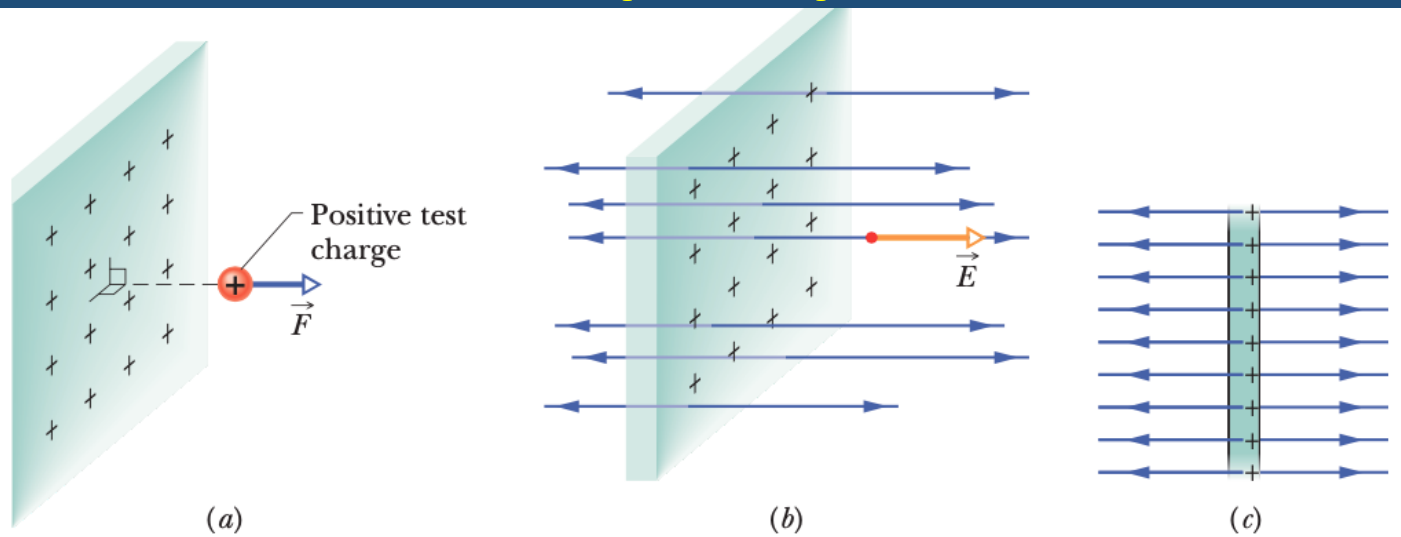
(a) The electrostatic force $\vec{F}$ acting on a positive test charge near a sphere of uniform negative charge.
(b) The electric field vector $\vec{E}$ at the location of the test charge, and the electric field lines in the space near the sphere. The field lines extend *toward* the negatively charged sphere. (They originate on distant positive charges.)

Chapter 2. Electric fields (2.0h)

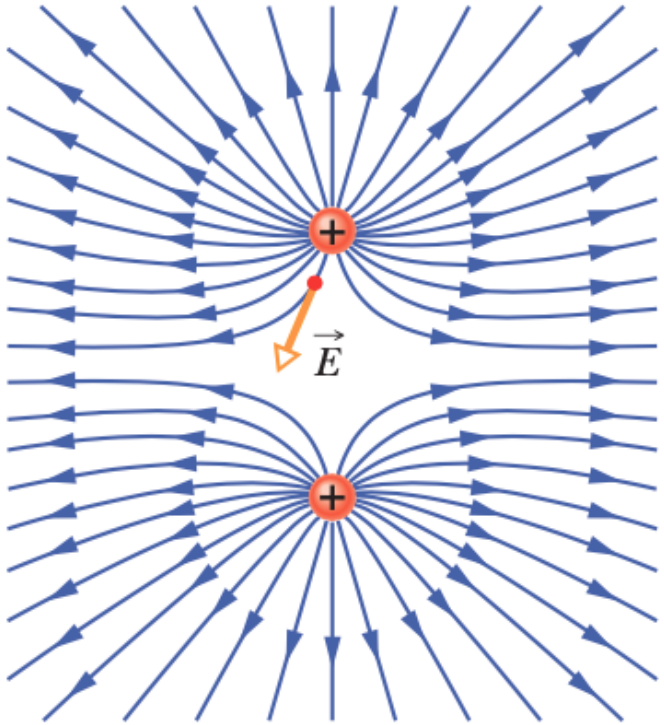
Electric field lines

Uniform electric field: the electric field has the same magnitude and direction at every point within the field.

Nonuniform electric field: there is variation from point to point.



(a) The force on a positive test charge near a very large, conducting sheet with uniform positive charge on one side.
(b) The electric field vector $\vec{E}$ at the test charge's location, and the nearby electric field lines, extending away from the sheet.
(c) Side view.

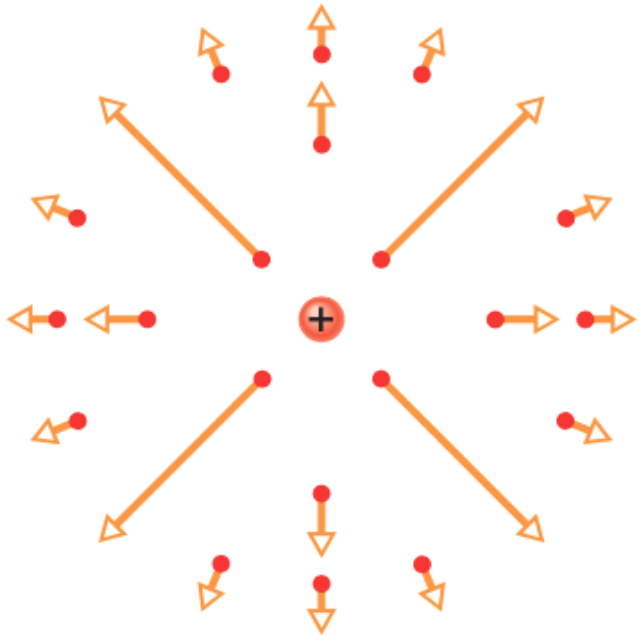


Field lines for two particles with equal positive charge. Doesn't the pattern itself suggest that the particles repel each other?

Chapter 2. Electric fields (2.0h

2.2 The electric field due to a charged particle

2.2.1 The electric field due to a point charge



The electric field vectors at various points around a positive point charge.

✓ The **electrostatic force** on the test charge q_0 at any point near the particle, at distance r due to the particle with charge q : (from Coulomb/s law)

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2}\hat{r}$$

is directly *away* from the particle if q is positive,
is directly *toward* the particle if q is negative.

✓ The **electric field** at the location of the test charge q_0, at distance r due to the particle with charge q :

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} \qquad (2.2)$$

- is directly *away* from the point charge if q is positive.
- is directly *toward* the point charge if q is negative.

✓ The magnitude of **electric field** at distance r due to the particle with charge q :

$$E = \frac{1}{4\pi\varepsilon_0}\frac{|q|}{r^2} \qquad (2.3)$$

Chapter 2. Electric fields (2.0h)

✓ The principle of superposition:

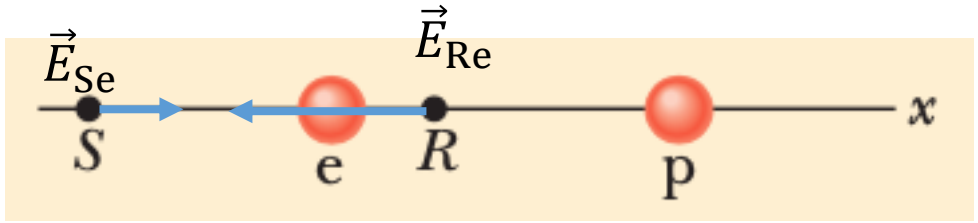
The net force by *n* charged particles acting on the test particle q_0 is given by the vector sum:

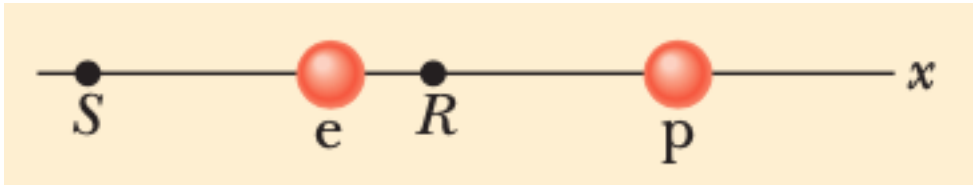
$$\vec{F}_0 = \vec{F}_{01} + \vec{F}_{02} + \vec{F}_{03} + \cdots + \vec{F}_{0n}$$

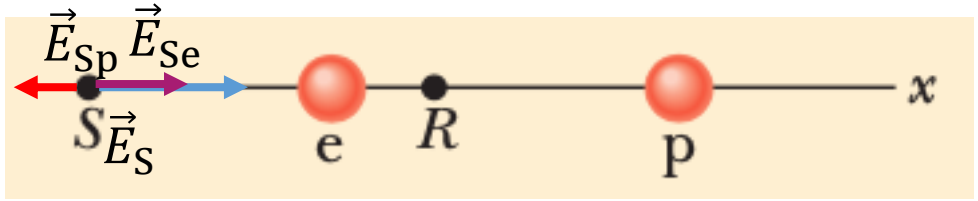
$\vec{F}_{0n}$: the force on the test charge q_0 due to the presence of particle *n*.

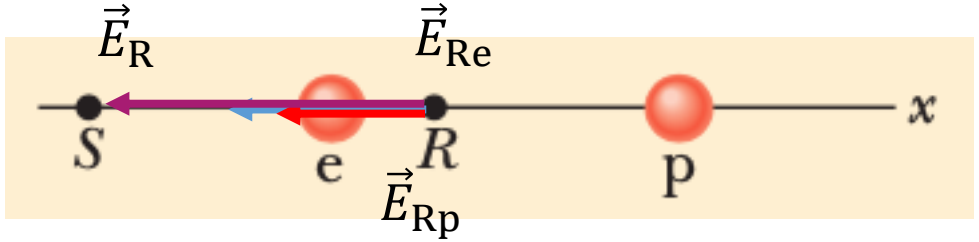
$$\vec{E} = \frac{\vec{F}_0}{q_0} = \frac{\vec{F}_{01}}{q_0} + \frac{\vec{F}_{02}}{q_0} + \frac{\vec{F}_{03}}{q_0} + \cdots + \frac{\vec{F}_{0n}}{q_0} = \vec{E}_1 + \vec{E}_2 \ldots + \vec{E}_n \qquad (2.4)$$

Checkpoint 2.1:
The figure shows a proton p and an electron e on an *x* axis. What is the direction of the electric field due to the electron at (a) point *S*? and (b) point *R*?
What is the direction of the net electric field at (c) point *R* and (d) point *S*?





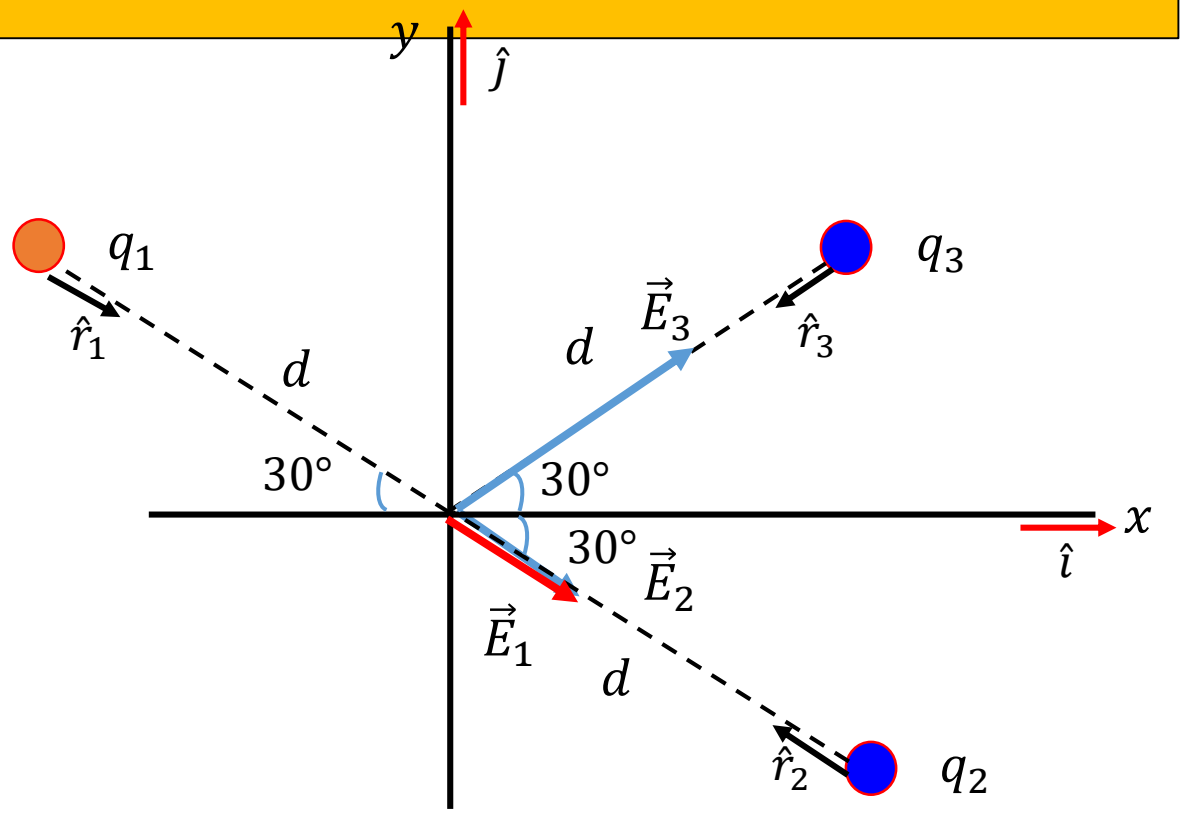




Chapter 2. Electric fields (2.0h)

Exp. 2.1: Net electric field due to three charged particles

Figure below shows three particles with charges $q_1 = +2Q$, $q_2 = -2Q$, $q_3 = -4Q$, each at a distance d from the origin. What net electric field $\vec{E}$ is produced at the origin?



The electric field $\vec{E}_n$ at the origin by particle *n*:

$$\vec{E}_n = \frac{1}{4\pi\varepsilon_0}\frac{q_n}{r_n^2}\hat{r}_n$$

Using the principle of superposition, the net electric field $\vec{E}$ produced at the origin is

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r_1^2}\hat{r}_1 + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{r_2^2}\hat{r}_2 + \frac{1}{4\pi\varepsilon_0}\frac{q_3}{r_3^2}\hat{r}_3$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{2Q}{d^2}\hat{r}_1 - \frac{1}{4\pi\varepsilon_0}\frac{2Q}{d^2}\hat{r}_2 - \frac{1}{4\pi\varepsilon_0}\frac{4Q}{d^2}\hat{r}_3$$

$$\vec{E} = \frac{2}{4\pi\varepsilon_0}\frac{2Q}{d^2}(\cos 30°\hat{\imath} - \sin 30°\hat{\jmath}) + \frac{1}{4\pi\varepsilon_0}\frac{4Q}{d^2}(\cos 30°\hat{\imath} + \sin 30°\hat{\jmath})$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{4Q}{d^2}(2\cos 30°\hat{\imath}) = \frac{1}{4\pi\varepsilon_0}\frac{4\sqrt{3}Q}{d^2}\hat{\imath}$$

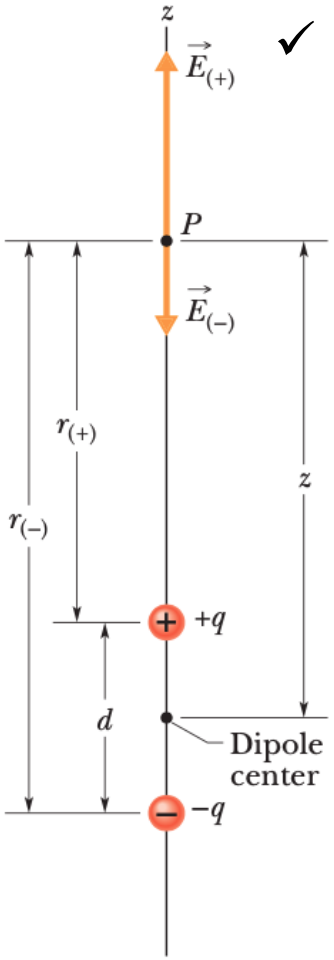
The net electric field $\vec{E}$ at the origin has magnitude of $\frac{1}{4\pi\varepsilon_0}\frac{4\sqrt{3}Q}{d^2}(\mathrm{N}/C)$ and the direction of the positive direction of the *x* axis.

Chapter 2. Electric fields (2.

2.3 The electric field due to a dipole

The electric field due to an electric dipole

✓ **Electric dipole**: two particles have the same magnitude q but opposite signs, are separated by distance d and lie along the dipole axis (z axis).



An electric dipole.

The electric field vectors $\vec{E}_{(+)}$ and $\vec{E}_{(-)}$ at point P on the dipole axis result from the dipole's two charges.

Point P is at distances $r_{(+)}$ and $r_{(-)}$ from the individual charges that make up the dipole.

✓ The magnitude of the net **electric field** at P:

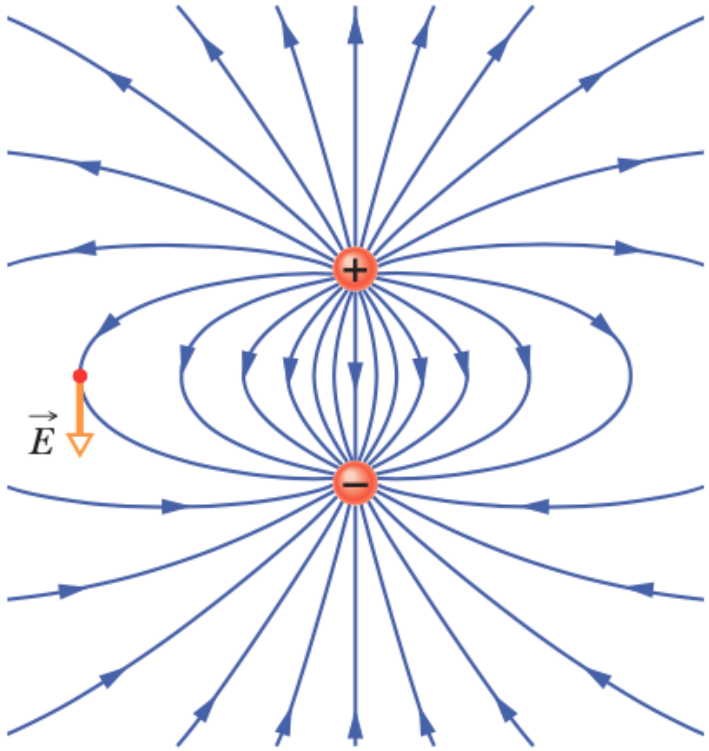
$$E = E_{(+)} - E_{(-)} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r_{(+)}^2} - \frac{1}{4\pi\varepsilon_0}\frac{q}{r_{(-)}^2}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{\left(z-\frac{d}{2}\right)^2} - \frac{1}{4\pi\varepsilon_0}\frac{q}{\left(z+\frac{d}{2}\right)^2} \qquad (2.5)$$

$$E = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{\left(z-\frac{d}{2}\right)^2} - \frac{1}{\left(z+\frac{d}{2}\right)^2}\right) = \frac{q}{4\pi\varepsilon_0 z^2}\frac{\frac{2d}{z}}{\left(1-\left(\frac{d}{2z}\right)^2\right)^2}$$

$$E = \frac{q}{2\pi\varepsilon_0 z^3}\frac{d}{\left(1-\left(\frac{d}{2z}\right)^2\right)^2} \qquad (2.6)$$

z: the distance from the dipole center to P



The pattern of electric field lines around an electric dipole, with an electric field vector $\vec{E}$ shown at one point (tangent to the field line through that point).

Chapter 2. Electric fields (2.0h)

The electric field due to an electric dipole

$$E = \frac{q}{2\pi\varepsilon_0 z^3}\frac{d}{\left(1-\left(\frac{d}{2z}\right)^2\right)^2} \qquad (2.6)$$

$z \gg d \Rightarrow \frac{d}{2z} \ll 1 \Rightarrow$ (2.6):

$$E = \frac{qd}{2\pi\varepsilon_0 z^3} \qquad (2.7)$$

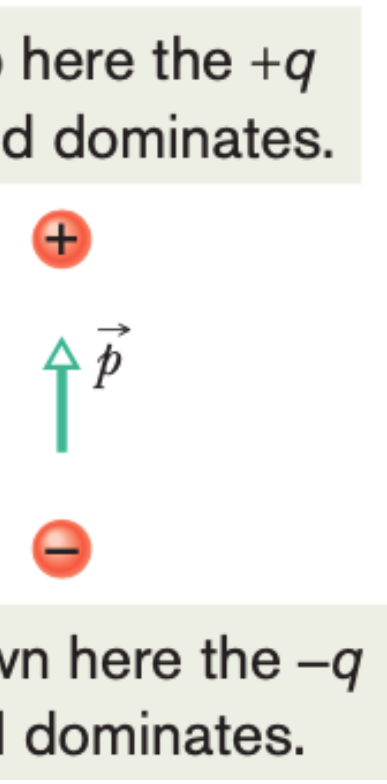
$p = qd$ **electric dipole moment,** unit (C.m) (2.8)

$\vec{p} = q\vec{d}$

The dipole moment $\vec{p}$ of the dipole points from the negative charge to the positive charge.

The **electric field** at *P* **by the dipole** (2.7):

$$E = \frac{1}{2\pi\varepsilon_0}\frac{p}{z^3} \qquad (2.9)$$

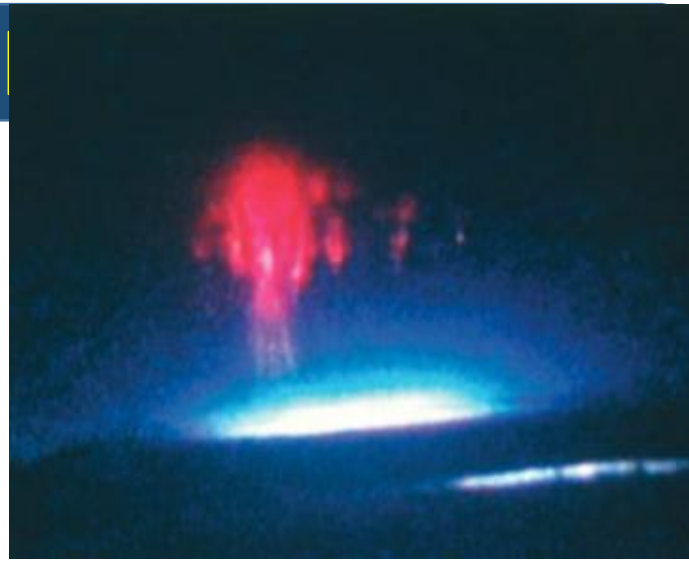


Chapter 2. Electric fields (2.0

Exp. 2.2: Electric dipole at atmospheric sprites

Figure (a) below shows sprites which are huge flashes occurring far above large thunderstorm. They were seen for decades by pilots flying at night, but they were so brief and dim that most pilots figured they were just illusions.

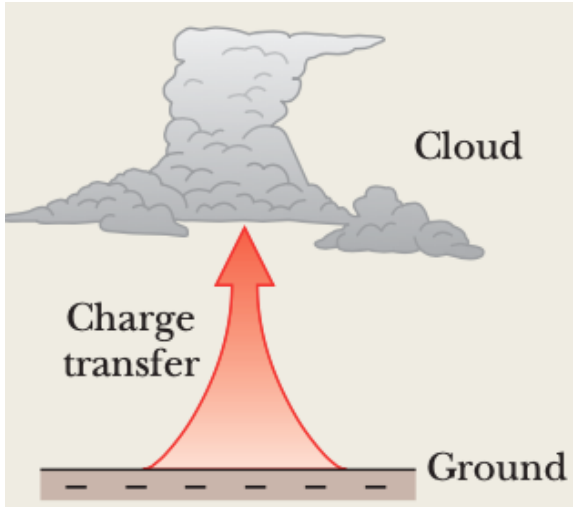
Then in the 1990s sprites were captured on video. They are still not well understood but are believed to be produced by when especially powerful lightning occurs between the ground and storm clouds, particularly when the lightning transfers a huge amount of negative charge $-q$ from the ground to the base (Figure (b)).



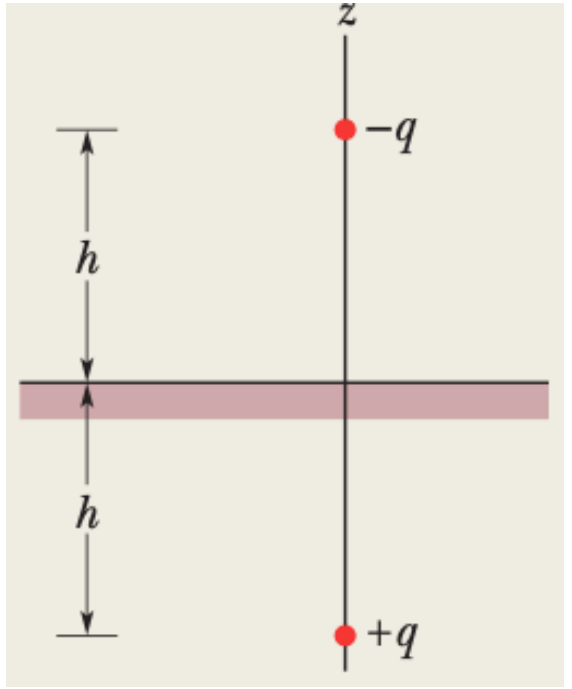
(a) Photograph of a sprite (NASA).

Just after such a transfer, the ground has a complicated distribution of positive charges. However, we can model the electric field due to the charges in the clouds and the ground by assuming a vertical electric dipole that has charge $-q$ at cloud height h and charge $+q$ at below-ground depth h (Figure (c)).

If $q = 200\ \text{C}$ and $h = 6\ \text{km}$, what is the magnitude of the dipole's electric field at altitude $z_1 = 30\ \text{km}$ somewhat above the clouds and altitude $z_2 = 60\ \text{km}$ somewhat above the stratosphere?



(b) Lightning in which a large amount of negative charge is transferred from ground to cloud base



(c) The cloud – ground system modeled as a vertical electric dipole.

Chapter 2. Electric fields (2.0h)

The magnitude of the dipole's electric field:

$$E = \frac{1}{2\pi\varepsilon_0}\frac{p}{z^3} = \frac{1}{2\pi\varepsilon_0}\frac{q2h}{z^3}$$

$2h$ is the separation between $+q$ and $-q$

At altitude $z_1 = 30$ km above the clouds

$$E_1 = \frac{1}{2\pi\varepsilon_0}\frac{q2h}{{z_1}^3}$$

$$= \frac{1}{2\pi 8.854\times10^{-12}}\frac{200\times2\times6\times10^3}{(3\times10^4)^3}$$

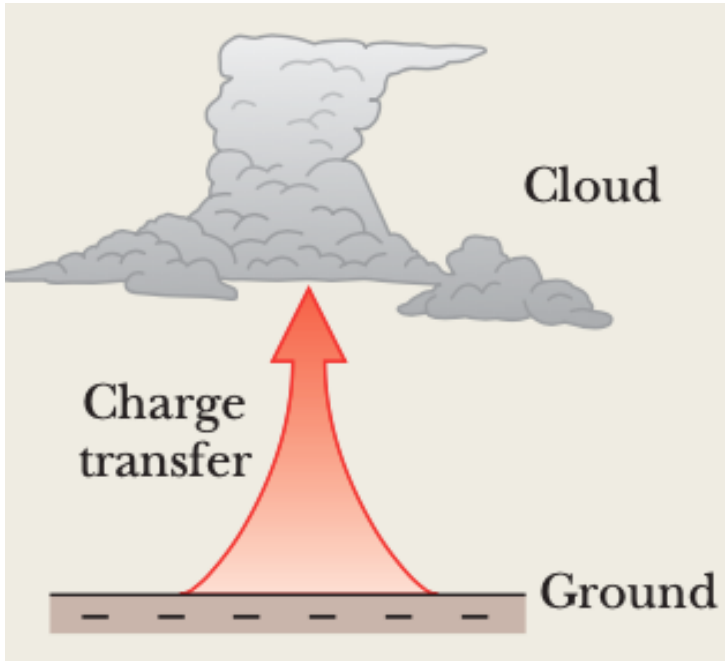
$$= 1.6\times10^3(\mathrm{N/C})$$

At altitude $z_2 = 60$ km above the stratosphere

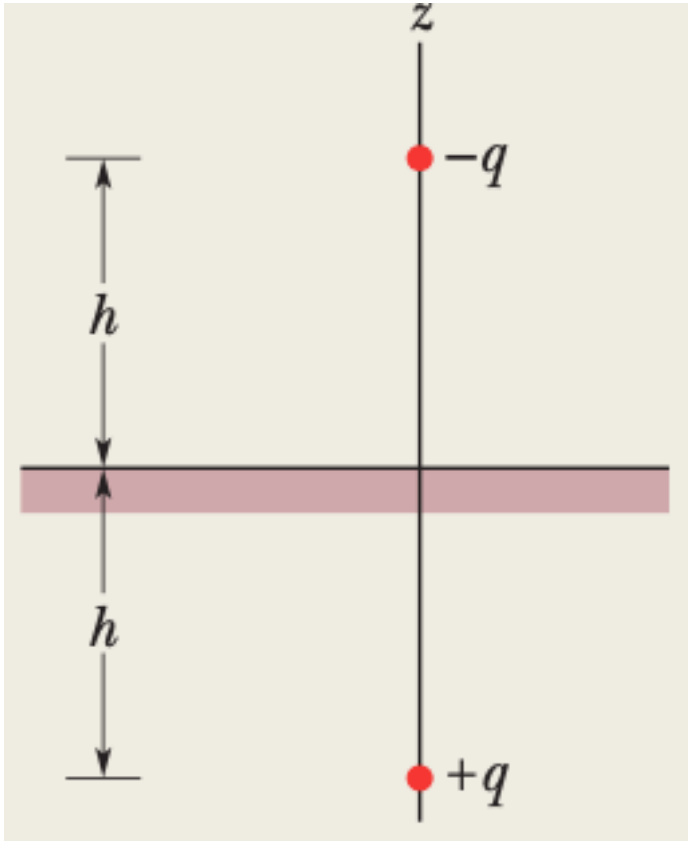
$$E_2 = \frac{1}{2\pi\varepsilon_0}\frac{q2h}{{z_2}^3}$$

$$= \frac{1}{2\pi 8.854\times10^{-12}}\frac{200\times2\times6\times10^3}{(6\times10^4)^3}$$

$$= 200(\mathrm{N/C})$$



(b) Lightning in which a large amount of negative charge is transferred from ground to cloud base



(c) The cloud – ground system modeled as a vertical electric dipole.

Chapter 2. Electric fields (2.0h)

2.4 The electric field due to a line of charge

The electric field due to a line of charge

$$\lambda = \frac{q}{2\pi R}$$

✓ A thin ring of radius R with a uniform distribution of positive charge along its circumference.

? The electric field at an arbitrary point *P* on the central axis (the axis through the ring's center and perpendicular to the plane of the ring), at distance z from the center point.

✓ **Linear charge density** λ (the charge per unit length, C/m)

Name	Symbol	SI Unit
Charge	q	C
Linear charge density	λ	C/m
Surface charge density	σ	C/m^2
Volume charge density	ρ	C/m^3

➢ Divide the ring into differential elements of charge dq which can be treated as particles.

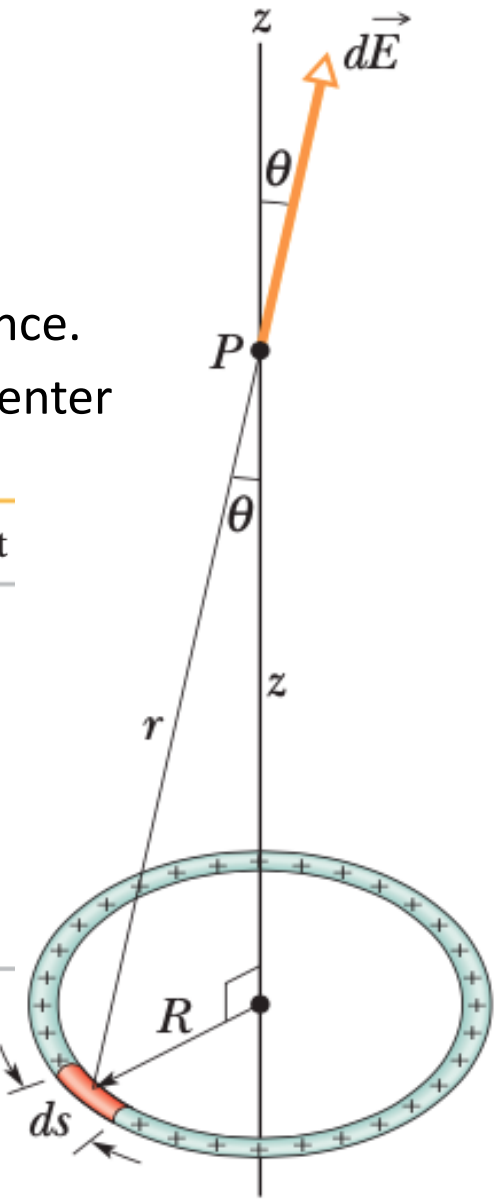
ds is the arc length of the dq element.

$$dq = \lambda ds \qquad (2.10)$$

➢ The differential electric field $d\vec{E}$ due to dq at *P*, at distance r from the element: radially directs away from dq

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda ds}{r^2} \qquad (2.11)$$

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{\lambda ds}{(R^2 + z^2)} \qquad (2.12)$$



A ring of uniform positive charge. A differential element of charge occupies a length ds (greatly exaggerated for clarity). This element sets up an electric field $d\vec{E}$ at point *P*.

$d\vec{E}$ leans at angle θ to the central axis (z axis).

The electric field due to a line of charge

✓ Consider the charge element of opposite side of the ring

=> $d\vec{E}$ leans at angle θ to the central axis (z axis).

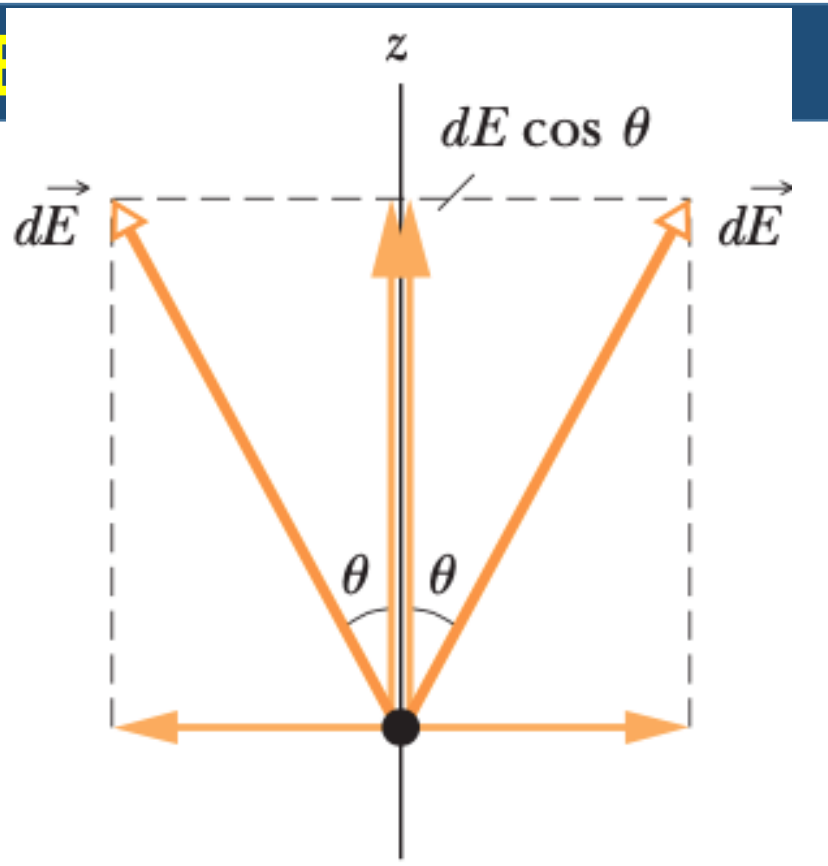
=> Two perpendicular components of $d\vec{E}$ cancel.

=> Two parallel components of $d\vec{E}$ add.

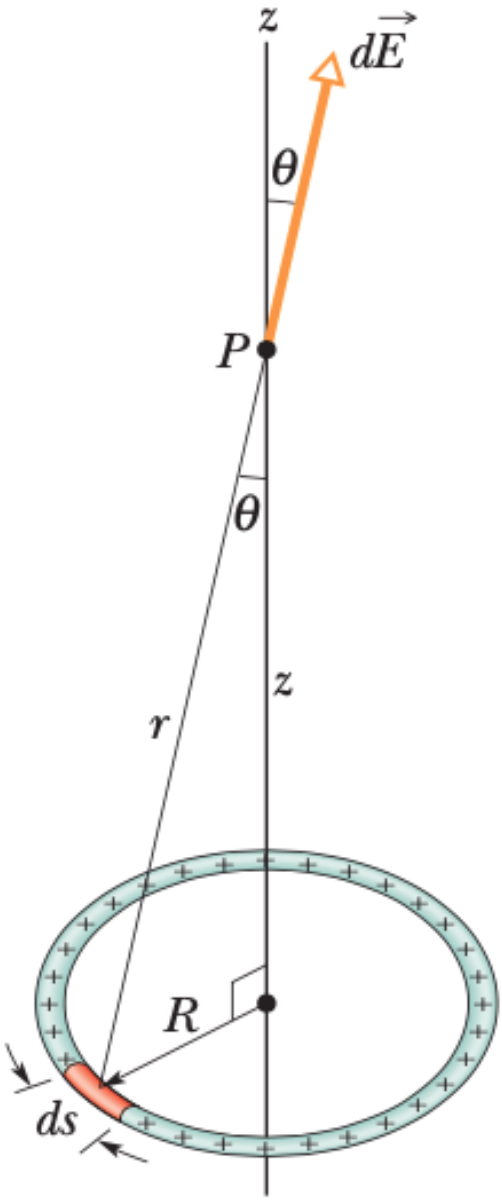
$$\cos\theta = \frac{z}{r} = \frac{z}{(R^2+z^2)^{1/2}} \qquad (2.13)$$

$$dE\cos\theta = \frac{1}{4\pi\varepsilon_0}\frac{\lambda ds}{(R^2+z^2)}\frac{z}{(R^2+z^2)^{1/2}}$$

$$dE\cos\theta = \frac{1}{4\pi\varepsilon_0}\frac{z\lambda ds}{(R^2+z^2)^{3/2}} \qquad (2.14)$$



The electric fields set up at *P* by a charge element and its symmetric partner (on the opposite side of the ring).
The components perpendicular to the z axis cancel; the parallel components add.



✓ Consider all the charge elements of the ring

$$E = \int dE\cos\theta = \frac{z\lambda}{4\pi\varepsilon_0(R^2+z^2)^{3/2}}\int_0^{2\pi R} ds = \frac{z\lambda 2\pi R}{4\pi\varepsilon_0(R^2+z^2)^{3/2}} \qquad (2.15)$$

Chapter 2. Electric fields (2.0h)

The electric field due to a line of charge

$$\lambda = \frac{q}{2\pi R}$$

$$E = \frac{zqR}{4\pi R\varepsilon_0 (R^2 + z^2)^{3/2}}$$

$$\boxed{E = \frac{zq}{4\pi\varepsilon_0 (R^2 + z^2)^{3/2}}} \qquad (2.17)$$

✓ If *P* is very far away the ring: $z \gg R$: $R^2 + z^2 \cong z^2$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{z^2} \qquad (2.18)$$

=> The ring "looks like" a point charge.

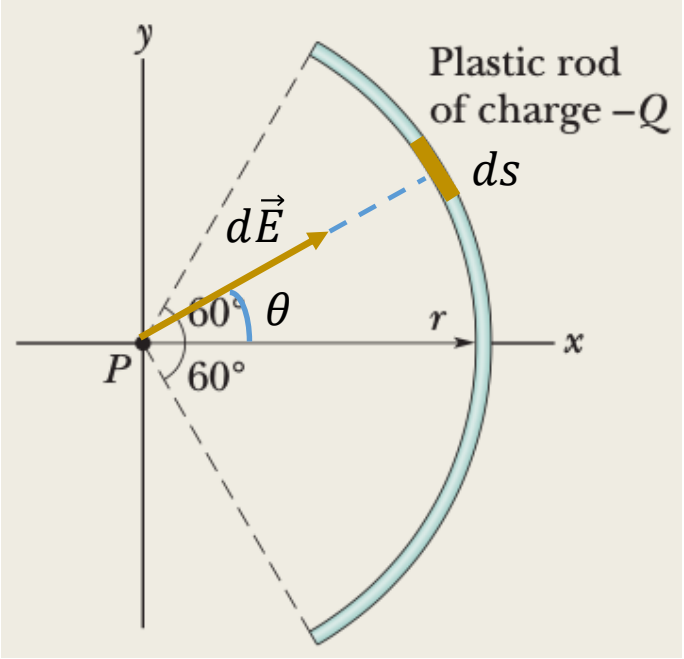
✓ If *P* is at the center the ring: $z = 0$

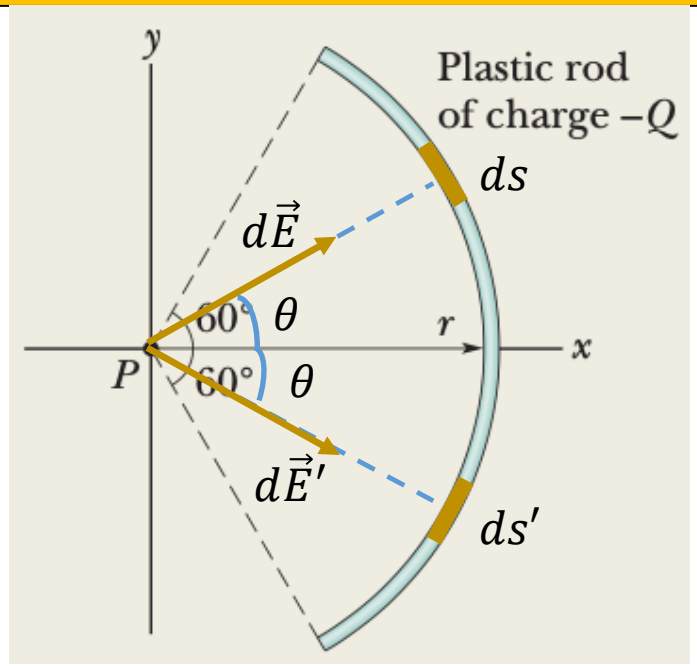
$$E = 0$$

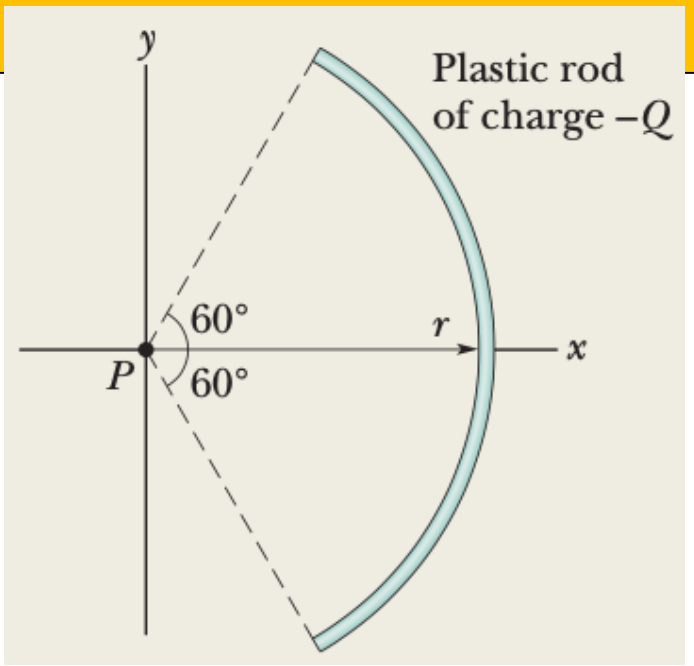
Chapter 2. Electric fields (2.0h)

Exp. 2.3: Electric field of a charged circular rod

Figure below shows a plastic rod with a uniform charge $-Q$. It is bent $120°$ circular arc of radius r and symmetrically paced across an *x* axis with the origin at the center of curvature *P* of the rod. In terms of Q and r, what is the electric field $\vec{E}$ due to the rod at point *P*?







- Divide the rod into differential elements of charge dq which can be treated as particles.

  ds is the arc length of the dq element. $dq = \lambda ds$

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda ds}{r^2}$$

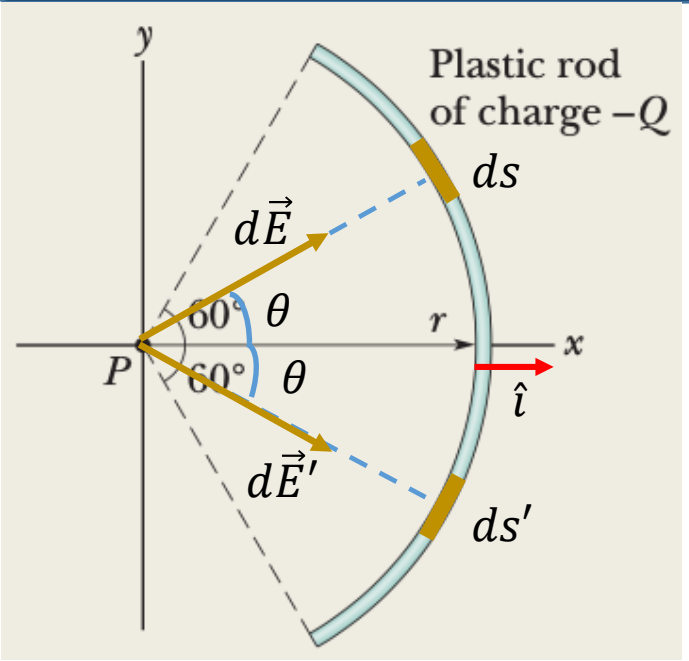
- ✓ Consider the charge element of opposite side of the arc

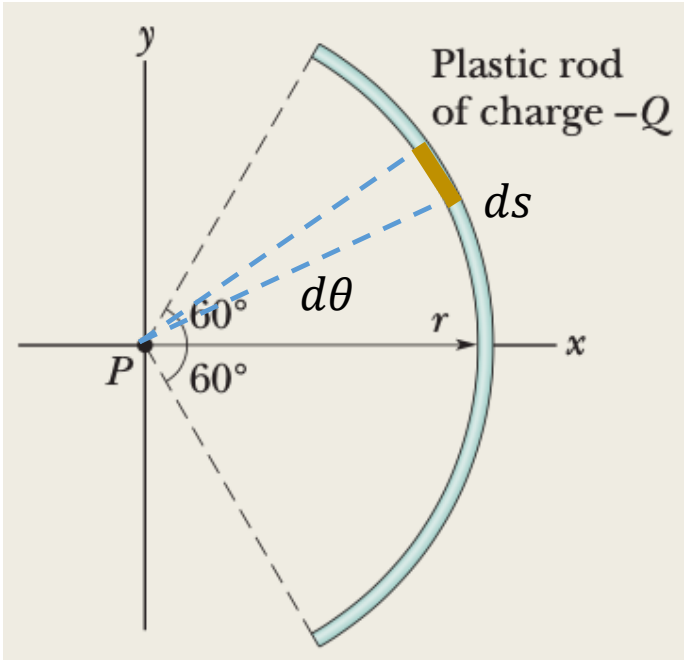
  => $\mathrm{d}\vec{E}$ leans at angle θ to x axis.

  => Two perpendicular components of $\mathrm{d}\vec{E}$ cancel.

  => Two parallel components of $\mathrm{d}\vec{E}$ add.

Chapter 2. Electric fields (2.0h)





$$\lambda = \frac{Q}{2\pi r \times \frac{1}{3}} = \frac{3Q}{2\pi r}$$

$$E = \frac{1.73\lambda}{4\pi\varepsilon_0 r} = \frac{1.73 \times 3Q}{4\pi\varepsilon_0 r \times 2\pi r} = \frac{0.83Q}{4\pi\varepsilon_0 r^2}$$

$$\vec{E} = \frac{0.83Q}{4\pi\varepsilon_0 r^2}\hat{\imath}$$

$$dE_x = dE\cos\theta = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\cos\theta ds}{r^2}$$

$$ds = rd\theta$$

$d\theta$ is the angle at point *P* that includes arc length ds.

$$E = \int dE\cos\theta = \frac{1}{4\pi\varepsilon_0}\frac{\lambda r}{r^2}\int_{-60°}^{60°}\cos\theta d\theta = \frac{1}{4\pi\varepsilon_0}\frac{\lambda}{r}\sin\theta\Big|_{-60°}^{60°} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\sqrt{3}}{r} = \frac{1.73\lambda}{4\pi\varepsilon_0 r}$$

Chapter 2. Electric fields (2.0h)

A field guide for lines of charge

For finding the electric field $\vec{E}$ produced at a point *P* by a line of uniform charge (circular or straight):

pick out an element dq of the charge
=> find $\mathrm{d}\vec{E}$ due to that element
=> integrate $\mathrm{d}\vec{E}$ over the entire line of charge.

- **Step 1**. If the line of charge is circular, let ds be the arc length of an element of the distribution.

If the line is straight, run an *x* axis along it and let dx be the length of an element. Mark the element on a sketch.

- **Step 2**. Relate the charge dq of the element to the length of the element with either $dq = \lambda ds$ or $dq = \lambda dx$. Consider dq and λ to be positive, even if the charge is actually negative.
- **Step 3**. Express the field $\mathrm{d}\vec{E}$ produced at *P* by dq in which $dq = \lambda ds$ or $dq = \lambda dx$.

If the charge on the line is positive, then at *P* draw a vector $d\vec{E}$ that points directly away from $\mathrm{d}q$.

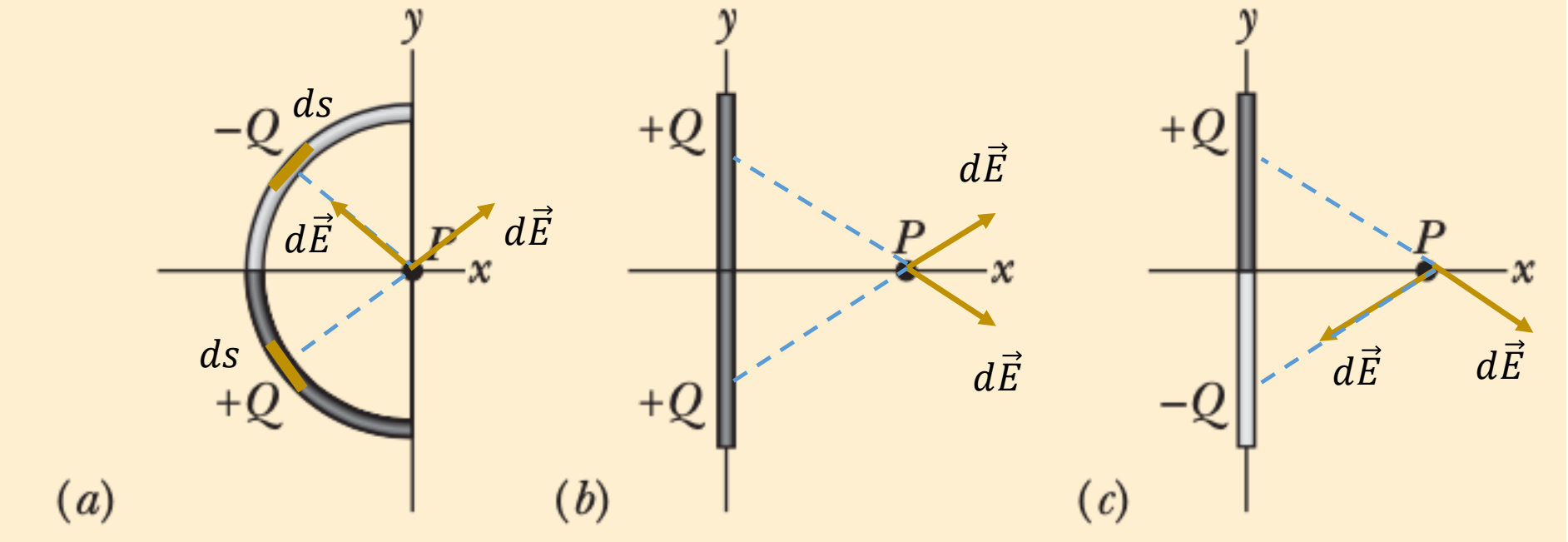
If the charge on the line is negative, then at *P* draw a vector $d\vec{E}$ that points directly toward $\mathrm{d}q$.

- **Step 4**. Look for any symmetry in the situation: canceling components or adding component.
- **Step 5**. Different types of uniform charge distributions:
  - ✓ Ring: point *P* on central axis of symmetry: $d\vec{E}$ in terms of θ; integrate over s.
  - ✓ Circular arc: point *P* at the center of curvature: $d\vec{E}$ in terms of θ; integrate over θ.
  - ✓ Straight line: point *P* on an extension of the line: $d\vec{E}$ in terms of x; integrate over x.
  - ✓ Straight line: point *P* at perpendicular distance y from the line: $\mathrm{d}\vec{E}$ in terms of x and y; integrate over x or y.
- **Step 6**. If the result s to be stated in terms of total charge Q of the distribution, replace λ with Q/L. (L: length of the distribution)

Chapter 2. Electric fields (2.0h)

Checkpoint 2.2:

The figure here shows three nonconducting rods, one circular and two straight. Each has a uniform charge of magnitude Q along its top half and another along its bottom half. For each rod, what is the direction of the net electric field at point *P*?



Chapter 2. Electric fields (2.0h)

2.5 The electric field due to a charged disk

Consider a circular plastic disk, with
radius R
a uniform surface charge density σ (charge per unit area) on its top surface.

? The electric field at an arbitrary point *P* on the central axis, at distance z from the center of the disk.

Superimpose a ring on the disk at an arbitrary radius $r \leq R$: as a charge element dq

The electric field component at *P* by this charge element dq:

$$dE = \frac{z dq}{4\pi\varepsilon_0 (r^2 + z^2)^{3/2}} \qquad (2.19)$$

$d\vec{E}$ points in the positive direction of the *z* axis.

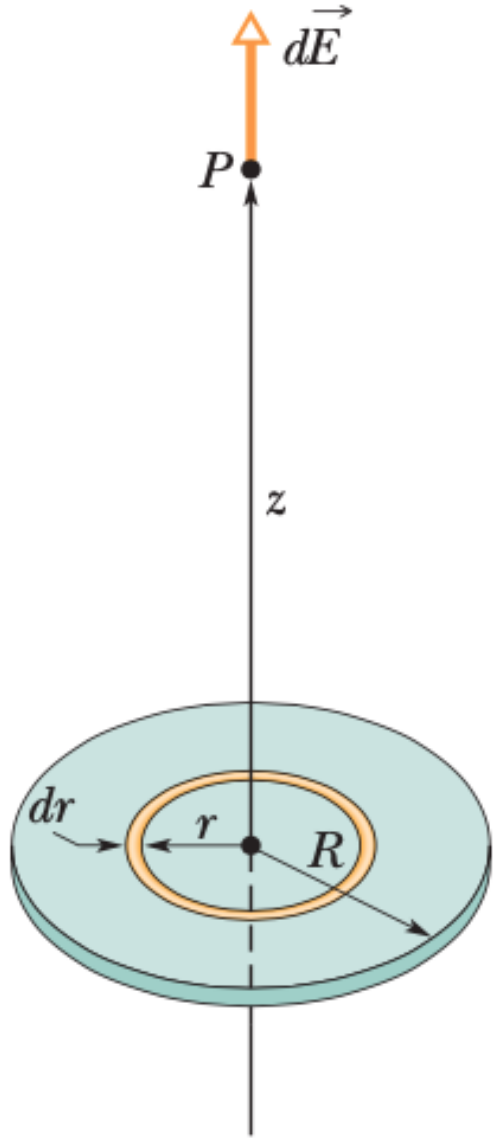
The total electric field at *P*: sum all dE by integrating (2.19) with respect to radius r

dr is the thickness of the ring.

$dA = (2\pi r)dr$ is the surface area of the ring.

$$dq = \sigma dA = \sigma(2\pi r)dr \qquad (2.20)$$

$$E = \int dE = \frac{\sigma z}{4\varepsilon_0} \int_0^R (r^2 + z^2)^{-3/2} 2r dr \qquad (2.21)$$



A disk of radius R and uniform positive charge. The ring shown has radius r and radial width dr. If sets up a differential electric field $d\vec{E}$ at point *P* on its central axis.

2.5 The electric field due to a charged disk

$$E = \frac{\sigma z}{4\varepsilon_0}\int_0^R (r^2+z^2)^{-3/2}2rdr$$

$$X = (r^2+z^2)$$

$$dX = 2rdr$$

$$E = \frac{\sigma z}{4\varepsilon_0}\int_{z^2}^{z^2+R^2} X^{-3/2}dX = \frac{\sigma z}{4\varepsilon_0}\frac{X^{-1/2}}{\frac{-1}{2}}\Bigg|_{z^2}^{z^2+R^2} = -\frac{\sigma z}{2\varepsilon_0}\left[(R^2+z^2)^{-1/2} - z^{-1}\right] \quad (2.22)$$

$$E = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{z}{\sqrt{R^2+z^2}}\right] \quad (2.23)$$

$R \to \infty, z$ finite: infinite sheet $\quad E = \frac{\sigma}{2\varepsilon_0}$ (2.24)

$z \to 0, R$ finite:
the point is very close to the disk $\quad E = \frac{\sigma}{2\varepsilon_0}$ (2.25)

Chapter 2. Electric fields (2.0h)

2.6 A point charge in an electric field

The electrostatic force $\vec{F}$ on a charged particle q surrounded by electric field $\vec{E}$:

$$\vec{F} = q\vec{E} \qquad (2.26)$$

Measuring the elementary charge



By Robert A. Millikan (American physicist) in 1910 - 1913.
Earned Nobel Prize (1923) in physics for this work.
Tiny oil drops are sprayed into chamber A, some of them become charged positively/negatively.

Consider a drop (negative charge q) that drifts downward through the small hole in conducting plate P_1 and into chamber C.

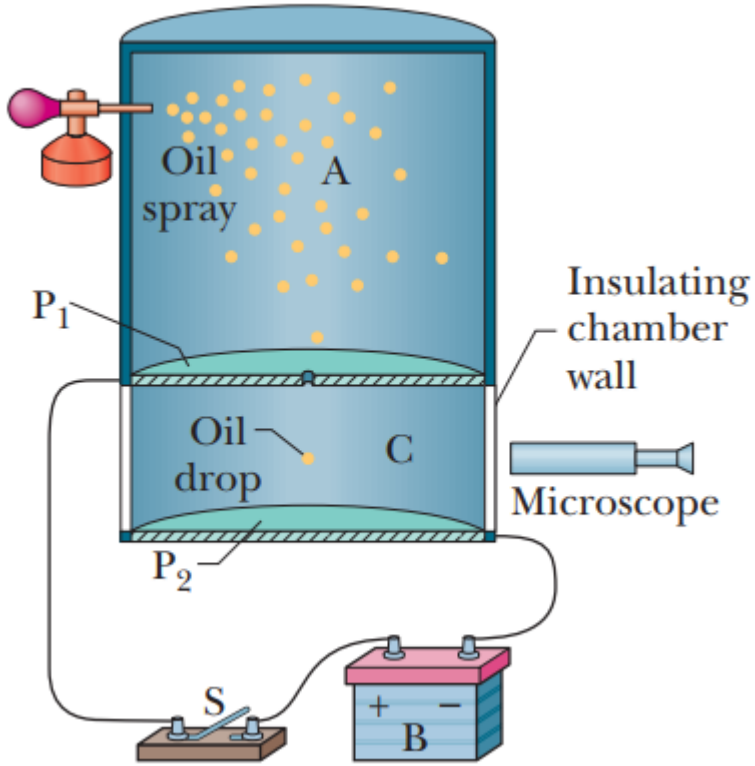
Switch S is open: battery B has no electrical effect on chamber C.

Switch S is closed: an excess positive charge on P_1 and an excess negative charge on P_2.

Downward-directed electric field $\vec{E}$ (↓) in chamber C.

=> Exert an electrostatic force $\vec{F}$ (↑) on any charged drop in chamber C.

By timing the motion oil drops with the switch opened/closed
=> Determining the effect of the charge q.



The Millikan oil-drop apparatus for measuring the elementary charge *e*. When a charged oil drop drifted into chamber C through the hole in plate P_1, its motion could be controlled by closing and opening switch S and thereby setting up or eliminating an electric field in chamber C. The microscope was used to view the drop, to permit timing of its motion.

https://www.youtube.com/watch?v=R0wOJhuBoCw

Chapter 2. Electric fields (2.0h)

2.6 A point charge in an electric field

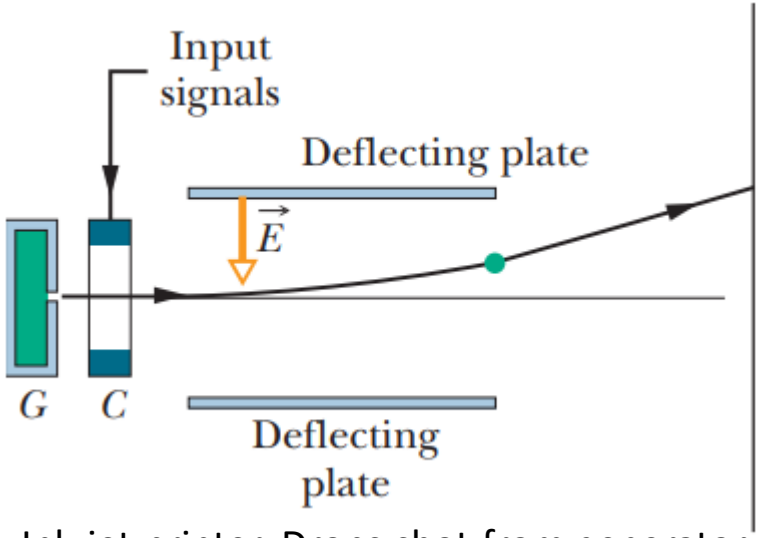
$$q = ne \qquad (2.27) \qquad n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

$e = 1.6 \times 10^{-19}(\mathrm{C})$ Elementary charge

Ink-Jet printing

Negatively charged drop moving between 2 conducting deflecting plates: uniform, downward-directed electric field $\vec{E}$ (↓).

=> Strikes at the paper.



Ink-jet printer. Drops shot from generator *G* receive a charge in charging unit C. An input signal from a computer controls the charge and thus the effect of field $\vec{E}$ on where the drop lands on the paper.

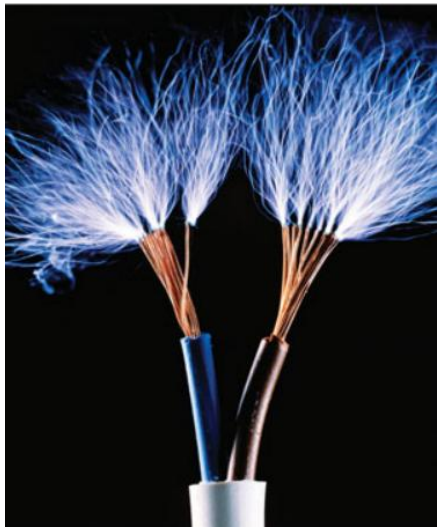
Electrical breakdown and sparking

If the magnitude of an electric field in air exceeds a certain critical value, E_C, the air undergoes ***electrical breakdown***.

Electrical breakdown: the field removes e from the atom in the air.

=> The air conducts electric current (the freed e are propelled into motion by the field).

=> Collide with atoms => These atoms emit light: "***sparks***"



The metal wires are so charged that the electric fields they produce in the surrounding space cause the air there to undergo electrical breakdown.

Chapter 2. Electric fields (2.0h)

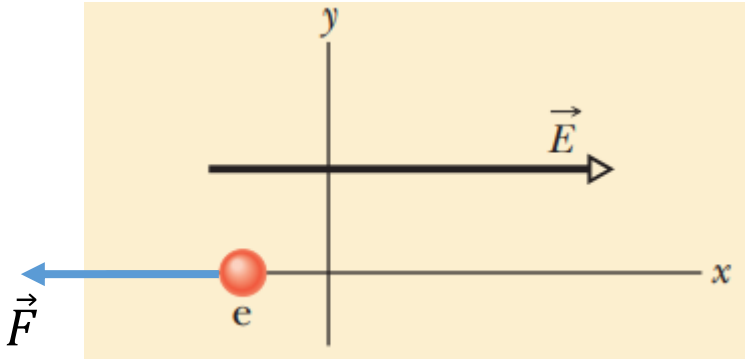
Checkpoint 2.3:

(a) In the figure, what is the direction of the electrostatic force on the e due to the external electric field shown?

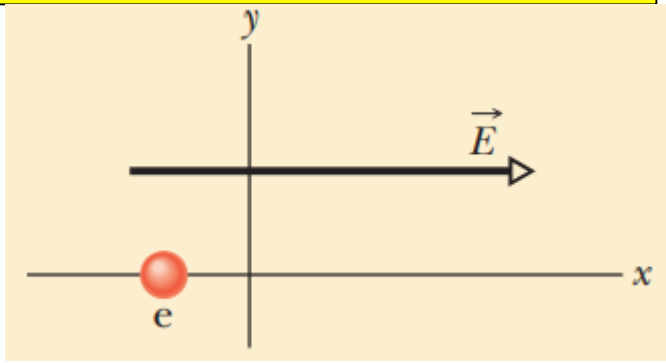
(b) In which direction will the e accelerate if it is moving parallel to the y axis before it encounters the external field?

(c) If, instead, the electron is initially moving rightward, will its speed increase, decrease, or remain constant?

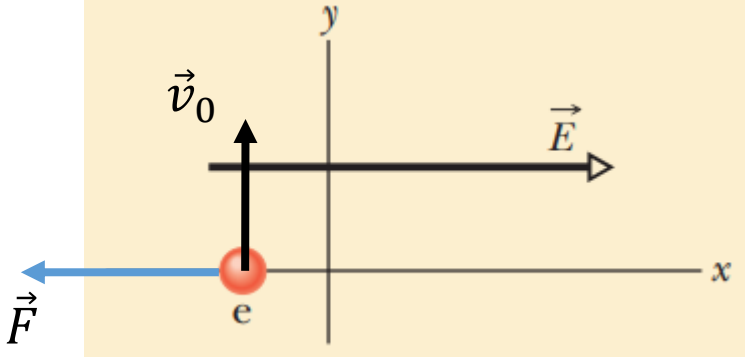
(a) The direction of the electrostatic force on the e due to the external electric field:



$$\vec{F} = -e\vec{E}$$

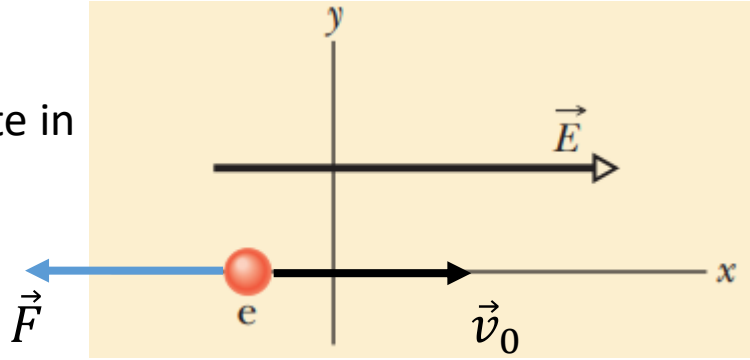


(b) If e is moving parallel to the y axis before it encounters the external field, it will accelerate in



x axis (in negative direction) in addition to *y* axis

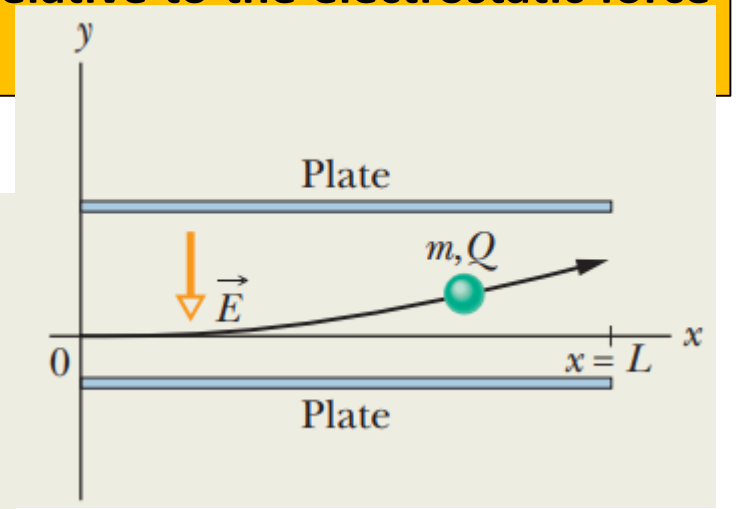
=> Parabolic trajectory.



(c) If the electron is initially moving rightward, its speed will decrease.

Exp. 2.4: Motion of a charged particle in an electric field

Figure below shows the deflecting plates of an ink-jet printer, with superimposed coordinate axes. An ink drop with a mass m of 1.3×10^{-10} kg and a negative charge of magnitude $Q = 1.5 \times 10^{-13}$ C enters the region between the plates, initially moving along the x axis with speed $v_x = 18$ m/s. The length L of each plates is 1.6 cm. The plates are charged and thus produce an electric field at all points between them. Assume that field $\vec{E}$ is downward directed, is uniform, and has a magnitude of 1.4×10^6 N/C . What is the vertical deflection of the drop at the far edge of the plates? (The gravitational force on the drop is small relative to the electrostatic force acting on the drop and can be neglected.)



The electrostatic force acting on the ink drop by the electric field:

$\vec{F} = Q\vec{E}$ is directly *upward*

$$\vec{a}_y = \frac{\vec{F}}{m} = \frac{Q\vec{E}}{m}$$

The vertical deflection $y = \frac{1}{2}a_y t^2 = \frac{1}{2}\frac{QE}{m}t^2$

The horizontal deflection $x = v_x t \Rightarrow t = \frac{x}{v_x}$

$$y = \frac{1}{2}\frac{QE}{m}t^2 = \frac{1}{2}\frac{QE}{mv_x^2}x^2$$

At the far edge of the plate $x = L$

$$y = \frac{1}{2}\frac{QE}{mv_x^2}L^2 = \frac{1}{2}\frac{1.5 \times 10^{-13} \times 1.4 \times 10^6}{1.3 \times 10^{-10} \times (18)^2}(1.6 \times 10^{-2})^2 = 6.4 \times 10^{-4}(\text{m}) = 0.64(\text{mm})$$

Chapter 2. Electric fields

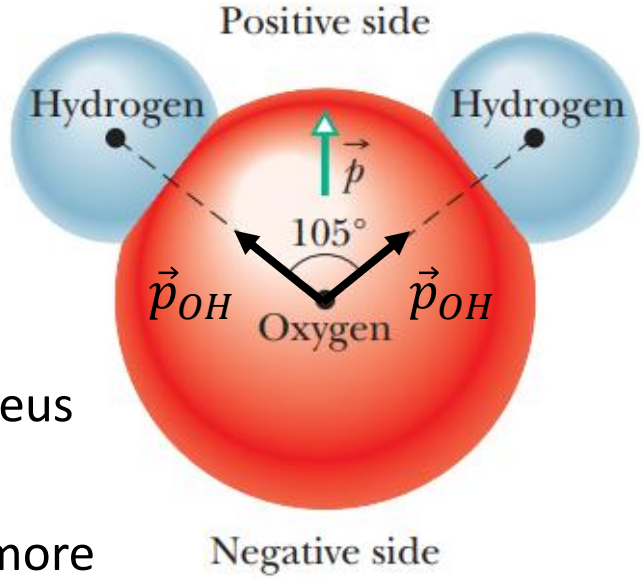
2.7 A point charge in an electric field

A dipole in an electric field

A molecule of H_2O:
1 oxygen nucleus (8 protons)
2 hydrogen nuclei (1 proton/each)

10 e tend to remain closer to the oxygen nucleus than to the hydrogen nuclei.

=> Oxygen side of the molecule slightly more negative than the hydrogen side of the molecule.

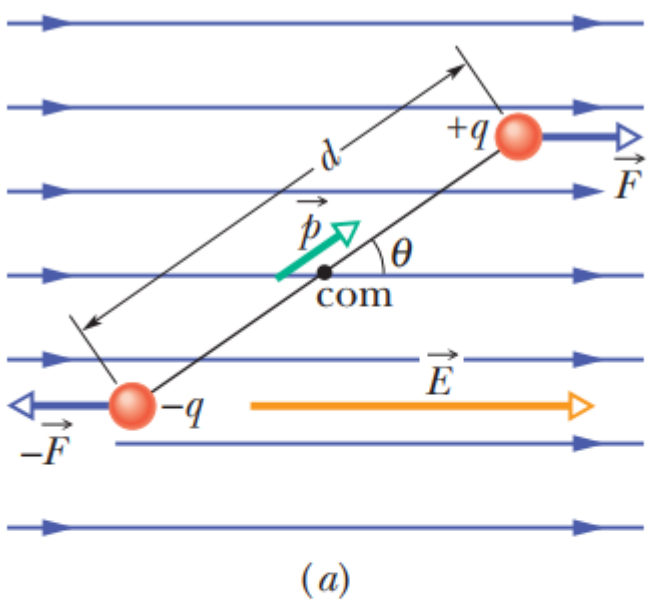


A molecule of H_2O, showing the three nuclei (represented by dots) and the regions in which the electrons can be located. The electric dipole moment $\vec{p}$ points from the (negative) oxygen side to the (positive) hydrogen side of the molecule.

Consider such an abstract dipole in a uniform external electric field $\vec{E}$: two centers of opposite charge, each of magnitude q, separated by a distance d.

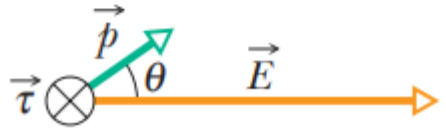
$(\vec{p}, \vec{E}) = \theta$

The net force $\vec{F} = 0$ $\vec{F} = q\vec{E}; -\vec{F} = -q\vec{E}$



(a)

The dipole is being torqued into alignment.



(b)

(a) An electric dipole in a uniform external field $\vec{E}$. Two centers of equal but opposite charge are separated by distance d. The line between them represents their rigid connection.

(b) Field $\vec{E}$ causes a torque $\vec{\tau}$ on the dipole. The direction of $\vec{\tau}$ is into the page, as represented by the symbol $\otimes$.

Chapter 2. Electric fields

2.7 A point charge in an electric field

A dipole in an electric field

The net torque

$$\tau = Fx\sin\theta + F(d-x)\sin\theta = Fd\sin\theta \qquad (2.28)$$

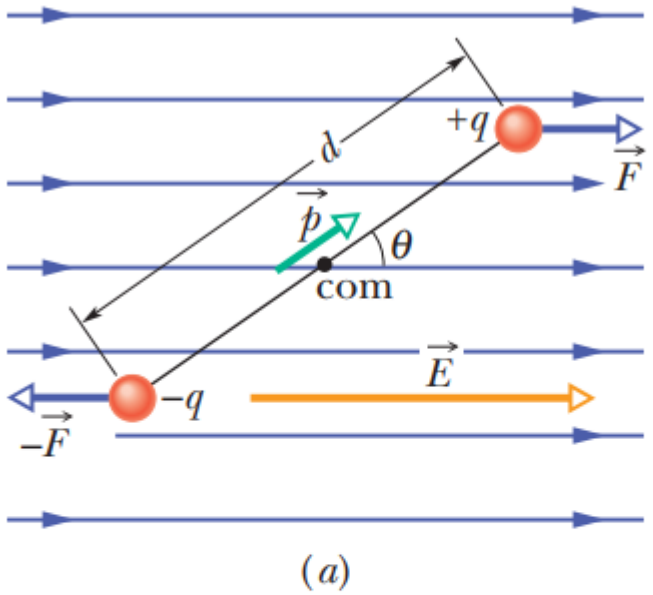
The magnitude of $\vec{\tau}$

$$\tau = pE\sin\theta \qquad (2.29)$$

$$\vec{\tau} = \vec{p} \times \vec{E} \qquad (2.30)$$

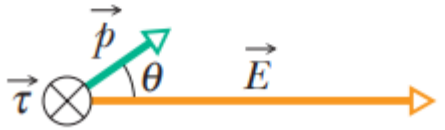
Clockwise rotation torque

$$\tau = -pE\sin\theta \qquad (2.31)$$



(a)

The dipole is being torqued into alignment.



(b)

(a) An electric dipole in a uniform external field $\vec{E}$. Two centers of equal but opposite charge are separated by distance d. The line between them represents their rigid connection.

(b) Field $\vec{E}$ causes a torque $\vec{\tau}$ on the dipole. The direction of $\vec{\tau}$ is into the page, as represented by the symbol ⊗.

Chapter 2. Electric fields (2.0h)

2.7 A point charge in an electric field

Potential energy of an electric dipole

Choose the potential energy to be zero when $\theta = 90°$

W: Work done by the field on the dipole
when the dipole is rotated from θ to $90°$

U: Potential energy at any angle θ

$$U = -W = -\int_{90°}^{0} \tau d\theta = \int_{90°}^{0} pE\sin\theta d\theta \qquad (2.32)$$

$$U = -pE\cos\theta \qquad (2.33)$$

$$U = -\vec{p} \cdot \vec{E} \qquad (2.34)$$

Work done on the dipole by the electric field when it rotates from an initial orientation θ_i to another orientation θ_f

$$W = -\Delta U = -\left(U_f - U_i\right) \qquad (2.35)$$

Work done on the dipole by the applied torque (*external agent*)

$$W_a = -W = \left(U_f - U_i\right) \qquad (2.36)$$

Chapter 2. Electric fields (2.0h)

2.7 A point charge in an electric field



Microwave cooking

Food contains water

Microwave oven: a rapidly oscillating electric field $\vec{E}$

=> Produce a torque on an electric dipole moment $\vec{p}$ to align $\vec{p}$ with $\vec{E}$

=> Water molecules continuously flip-flop
in a frustrated attempt to align with $\vec{E}$

=> Energy is transferred from the electric field to
the thermal energy of the water and of the food.

=> The flip-flop breaks some of the bonds

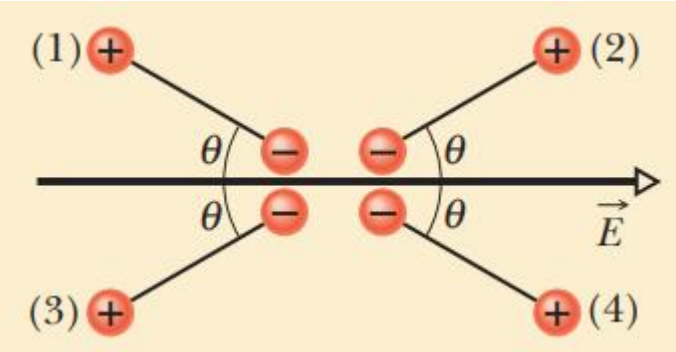
=> Energy is transferred to the random motion of the group and
then to the surrounding molecules

=> The thermal energy of the water is enough to cook the food.

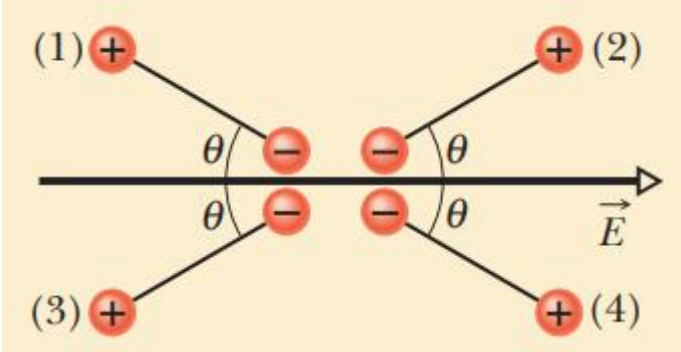
Chapter 2. Electric fields (2.0h)

Checkpoint 2.4:
The figure shows four orientations of an electric dipole in an external electric field. Rank the orientations according to (a) the magnitude of the torque on the dipole, (b) the potential energy of the dipole, greatest first.

(a) Rank the orientations according to the magnitude of the torque on the dipole



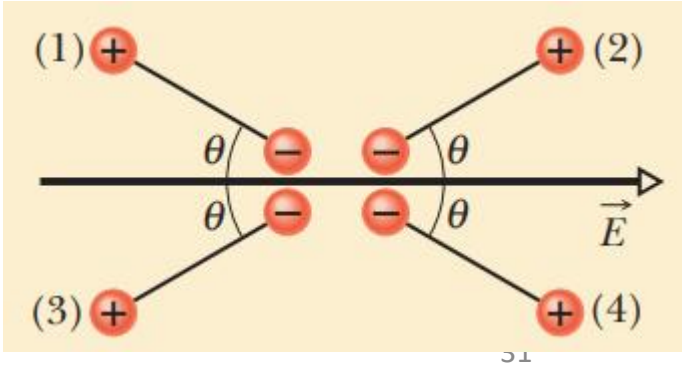
$$\vec{\tau} = \vec{p} \times \vec{E}$$



(4) – (1) – (2) – (3)

(b) Rank the orientations according to the potential energy of the dipole, greatest first.

$$U = -\vec{p} \cdot \vec{E}$$



(1) – (3) – (2) – (4)

Chapter 2. Electric fields (2.0h)

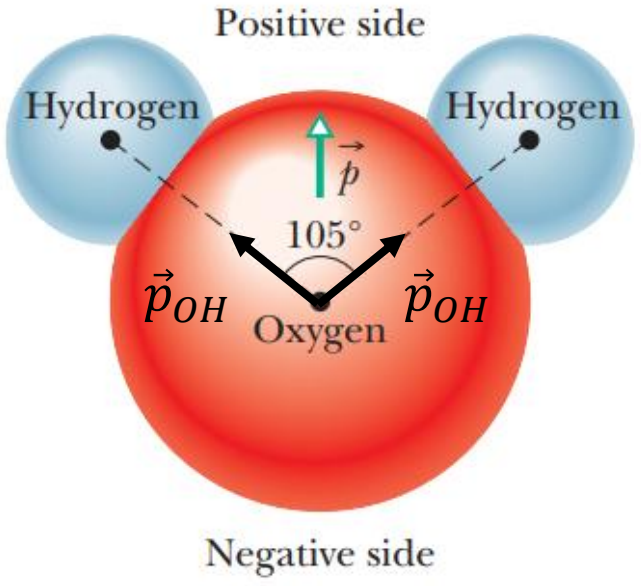
Exp. 2.5: Torque and energy of an electric dipole in an electric field

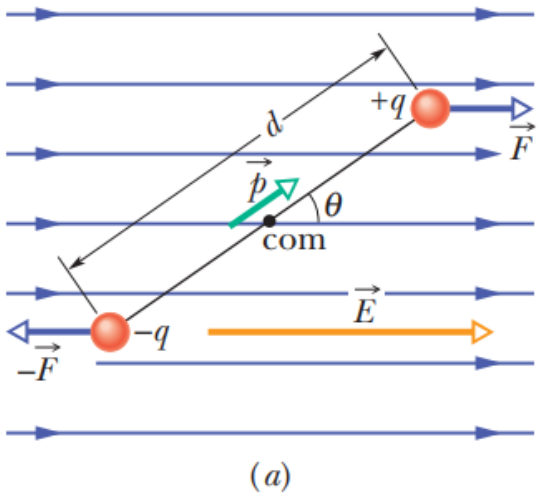
A neutral water molecule (H_2O) in its vapor state has an electric dipole moment of magnitude 6.2×10^{-30} Cm .

(a) How far apart are the molecule's centers of positive and negative charge?

(b) If the molecule is placed in an electric field of 1.5×10^4 N/C, what maximum torque can the field exert on it? (Such a field can easily be set up in the laboratory.)

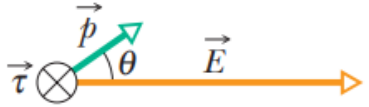
(c) How much work must an *external agent* do to rotate this molecule by $180°$ in this field, starting from its fully aligned position, for which $\theta = 0°$?





(a)

The dipole is being torqued into alignment.



(b)

$$p = qd \qquad d = \frac{p}{q} = \frac{p}{10e} = \frac{6.2 \times 10^{-30}}{10 \times 1.6 \times 10^{-19}}$$
$$= 3.9 \times 10^{-12}(\text{m}) = 3.9\ (\text{pm})$$

Maximum torque $\theta = 90°$

$$\tau_{\max} = pE\sin\theta$$
$$= 6.2 \times 10^{-30} \times 1.5 \times 10^4 \times \sin 90°$$
$$= 9.3 \times 10^{-26}(\text{Nm})$$

$$W_a = -W = (U_{180°} - U_0)$$
$$= (-pE\cos 180°) - (pE\cos 0) = 2pE$$
$$= 2 \times 6.2 \times 10^{-30} \times 1.5 \times 10^4$$
$$= 1.9 \times 10^{-25}(\text{J})$$

“When you really want something the whole universe conspires in helping you to achieve it”
(The Alchemist, Paulo Coelho)

TRƯỜNG ĐẠI HỌC KHOA HỌC VÀ CÔNG NGHỆ HÀ NỘI
UNIVERSITY OF SCIENCE AND TECHNOLOGY OF HANOI

GOOD LUCK and THANK YOU

Make it Simple but Significant!